Quality Information Document

Ocean Monitoring Indicator (OMI) of the Copernicus Marine Environment Monitoring Service

Arctic Sea Ice Extent

Issue: 3.0

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CHANGE RECORD

When the quality of the products changes, the QuID is updated and a row is added to this table. The third column specifies which sections or sub-sections have been updated. The fourth column should mention the version of the product to which the change applies.

Issue	Date	§	Description of Change	Author	Validated By
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I INTRODUCTION

I.1 Product name

ARTIC_OMI_SI_extent

I.2 Indicator evaluation method

Sea ice extent is defined here as the measurement of the area of the ocean covered at least by 15% of sea ice. Monthly mean (1993-2014) seasonal cycle time series of total area for sea ice extent are calculated for the northern hemisphere, e.g. Arctic Ocean and Nordic seas (excluding ice lakes). Total area monthly time series 1993-2023 for sea ice extent are calculated for the same domain. Sea Ice Extent is expressed here in units of millions of km2.

I.3 CMEMS products used as input for the computations

Product name (and Type)	Documentation
GLOBAL_MULTIYEAR_PHY_ENS_001_031	QUID : CMEMS-GLO-QUID-001-031.pdf PUM : CMEMS-GLO-PUM-001-031.pdf (copernicus.eu)

II SCIENTIFIC VALIDATION

More details and full scientific evaluation can be found in the CMEMS Ocean State Report (Samuelsen et al., 2016; Samuelsen et al., 2018).

The quality evaluation of ARCTIC_OMI_SI_extent is based on the "multi-product" approach as introduced in the second issue of the Ocean State Report (CMEMS OSR, 2017). Five global products have been used to build the ensemble mean, and its associated ensemble spread (Figure 1, dark red line and blue shaded area). Results of Fig. 1 indicate how different products compare to each other in view of their mean seasonal cycle and their interannual variability, e.g. annual means.

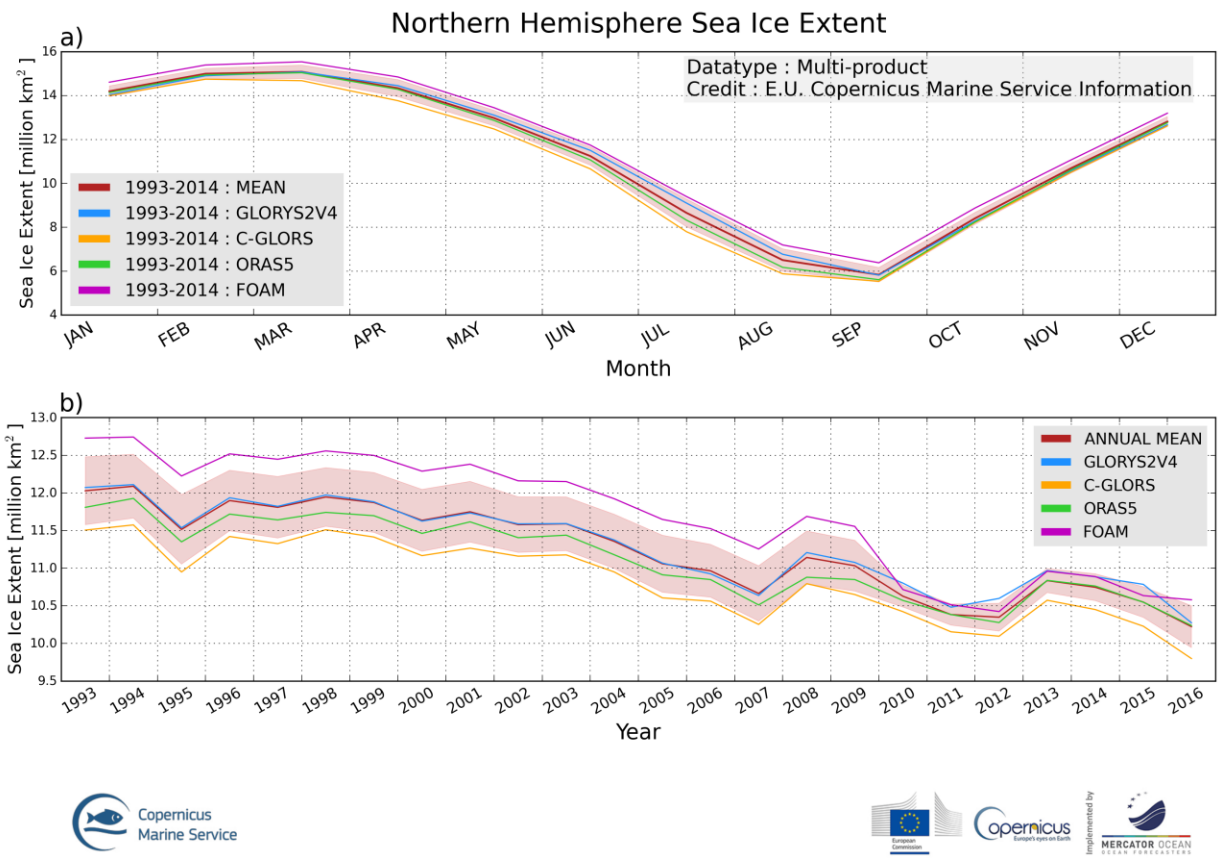

Figure 1: Arctic sea ice extent monthly means (Top panel) and annual mean sea ice extent (Bottom panel). Units are in millions of km2. Time series are based on different data products, i.e. global ocean reanalysis GLORYS (blue), C-GLORS (orange), ORAS5 (green), FOAM (purple) and the ensemble mean (red) and the ensemble spread (light blue). Details on the products are given in the corresponding PUM for this OMI.

III REFERENCES

A. Samuelsen et al., 2016: Sea Ice In: The Copernicus Marine Environment Monitoring Service Ocean State Report, issue 1, Journal of Operational Oceanography, Volume 9, 2016 - Issue sup2: The Copernicus Marine Environment Monitoring Service Ocean, http://dx.doi.org/10.1080/1755876X.2016.1273446.

A. Samuelsen et al., 2018: Sea Ice. In: The Copernicus Marine Environment Monitoring Service Ocean State Report, issue 2, Journal of Operational Oceanography.